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SEAT No. :

PE99

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[6579]-404

T.E. (Electronics and Computer Engineering) (Insem)

DATABASE MANAGEMENT SYSTEMS

(2019 Pattern) (Semester - I) (310341)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

Q1) a) What is file management system? What are drawbacks of file system? Compare file system and Database Management System. **[8]**

b) What is Database and Database Management System? What is purpose of database management systems? List the applications of DBMS. **[7]**

OR

Q2) a) Explain Primary key and Foreign key with example. Write difference between Primary key and Foreign Key. **[8]**

b) What is entity? Explain strong entity and weak entity with example. Compare strong entity and weak entity. **[7]**

Q3) a) What is Relational Model? Explain basic Structure of Relational Model. **[5]**

b) State and explain any 5 CODD's rules for RDBMS. **[5]**

c) Explain properties of relational database model. State the advantages of relational database model. **[5]**

OR

Q4) a) Define integrity constraints in relational database? Explain following constraints of relational database. **[5]**

i) Domain constraints

ii) Key Constraints or Uniqueness Constraints

b) Define functional dependency. Describe Single-valued Functional Dependency and Multi-valued Functional Dependency with suitable example. **[5]**

c) What is Normalization? State and explain 2NF & 3NF with example. **[5]**

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